



Big Data System Design (MBAI 5110G) Winter-2026

Week 8: Big Data Analytics

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Outline

- Big Data Analytics Terminologies
- Big Data Analytics Concepts and Analytics Types
- Analytics Life Cycle
- Big Data Analytics Techniques
- Semantic and Visual Analytics
- Big Data Business Intelligence
- Big Data Real-time Analytical Processing
- Integrating Enterprise Data Warehouse with Big Data Technologies

Terminologies of Bid Data Analytics

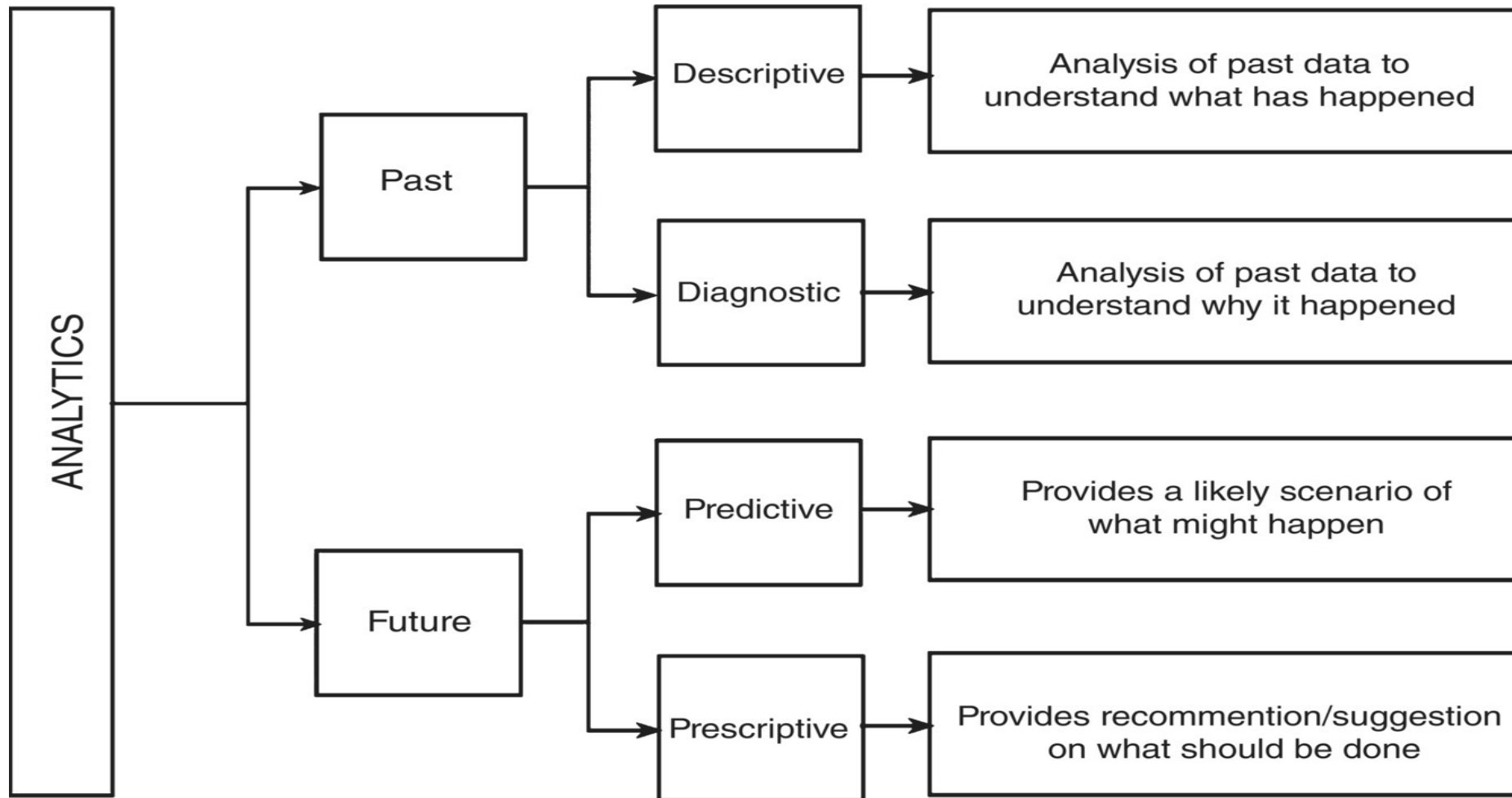
- **Data Warehouse/Enterprise Data Warehouse (EDW):**
 - A Data Warehouse is a centralized repository that stores data collected from multiple sources within an organization
 - Enables data analysts to perform reporting, analysis, and strategic decision-making efficiently
- **Business Intelligence (BI):**
 - Business Intelligence is the process of analyzing data to support better business decisions
 - Provides dashboards, reports, and visual insights for decision-making
- **Data Analytics:**
 - Data Analytics is the process of examining raw data to discover patterns, trends, and insights
 - Focuses on deep analysis & future prediction

Category /Terminologies	Main Purpose	Example Tools
Data Warehouse	Store & organize integrated data	Snowflake, BigQuery, Redshift
BI	Visualization & reporting	Power BI, Tableau, Qlik
Analytics	Deep data analysis & modeling	Python, R, Spark, SAS

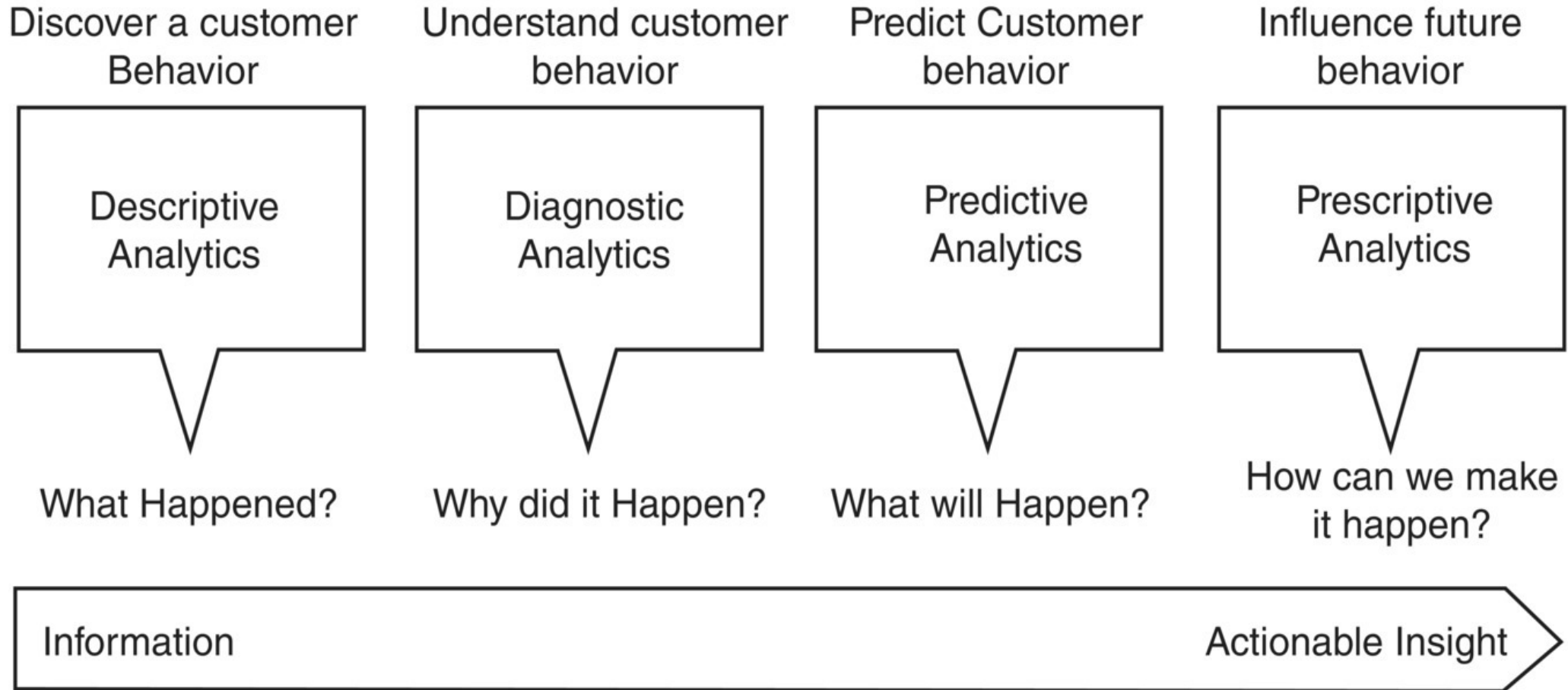
Big Data Analytics

- **Big Data Analytics** is the science of examining or analyzing large data sets with a variety of data types (structured, semi-structured, or unstructured data), which may be streaming or batch data
- **Purpose:**
 - Make better decisions
 - Discover new business opportunities
 - Improve performance and efficiency
 - Reduce operational cost by using advanced data analytics techniques
 - Compete effectively in the market
- **Applications:**
 - Marketing (response modeling, retention modeling)
 - Risk management (credit risk, operational risk, fraud detection)
 - Government sector (money laundering, terrorism detection)
 - Web (social media analytics) and many more

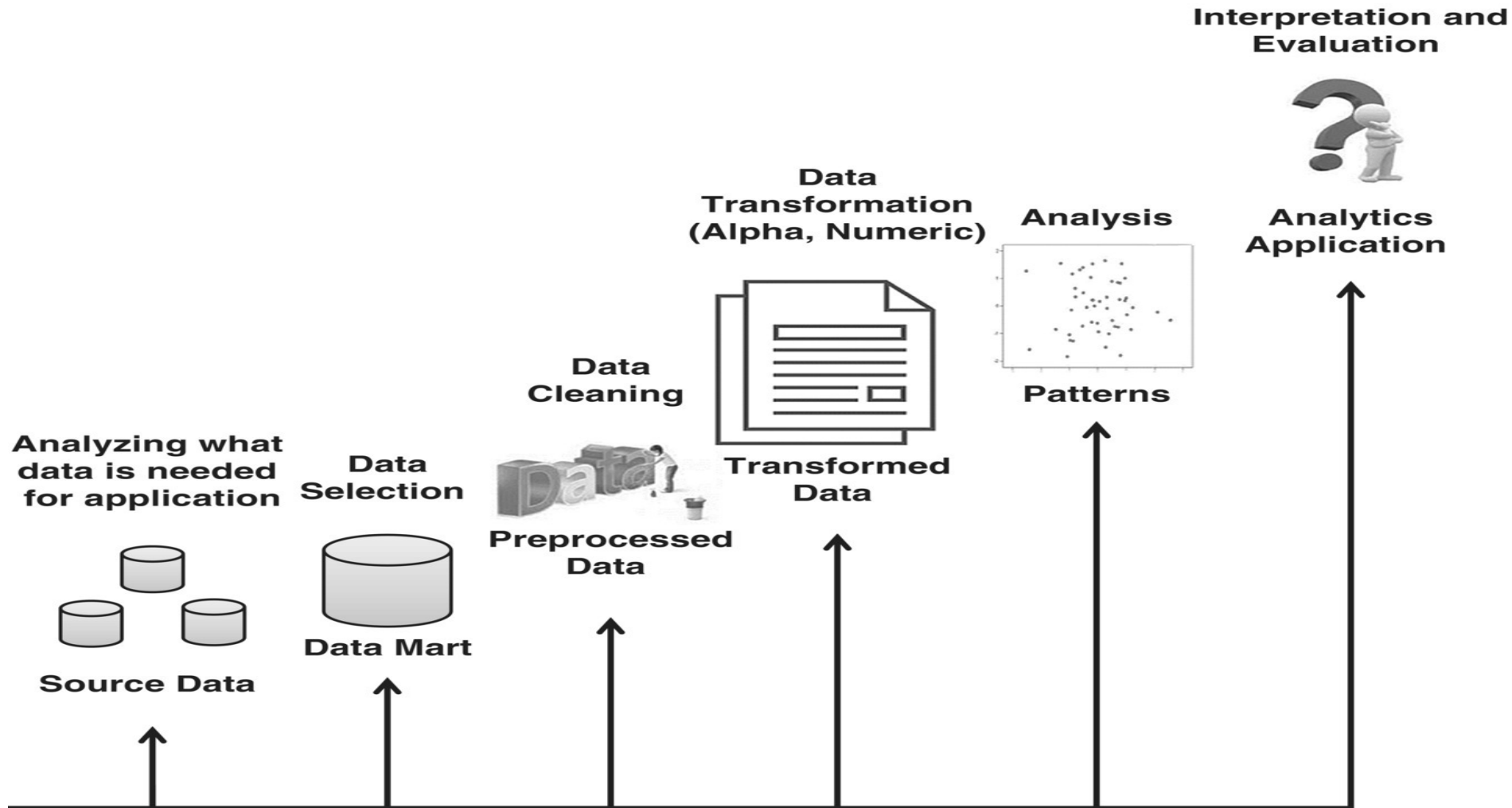
Types of Analytics



Types of Analytics (Cont.)

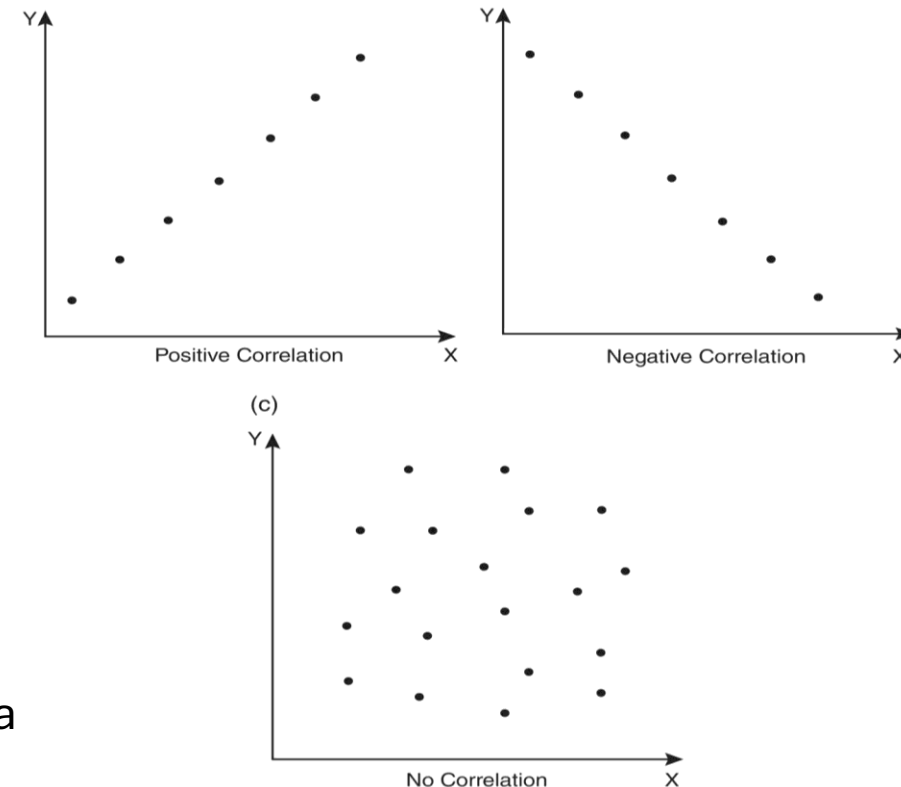


Analytics Lifecycle



Big Data Analytics Techniques

- Quantitative analysis
- Qualitative analysis
- Statistical analysis
 - **A/B Testing:**
 - It is also called split testing or bucket testing
 - It compares two versions of an object of interest to determine which among the two versions performs better.
 - **Correlation:**
 - Relationship between two variables
 - **Regression:**
 - Relationship between a dependent variable and an independent variable
 - The difference between regression and correlation is that correlation does not imply causation

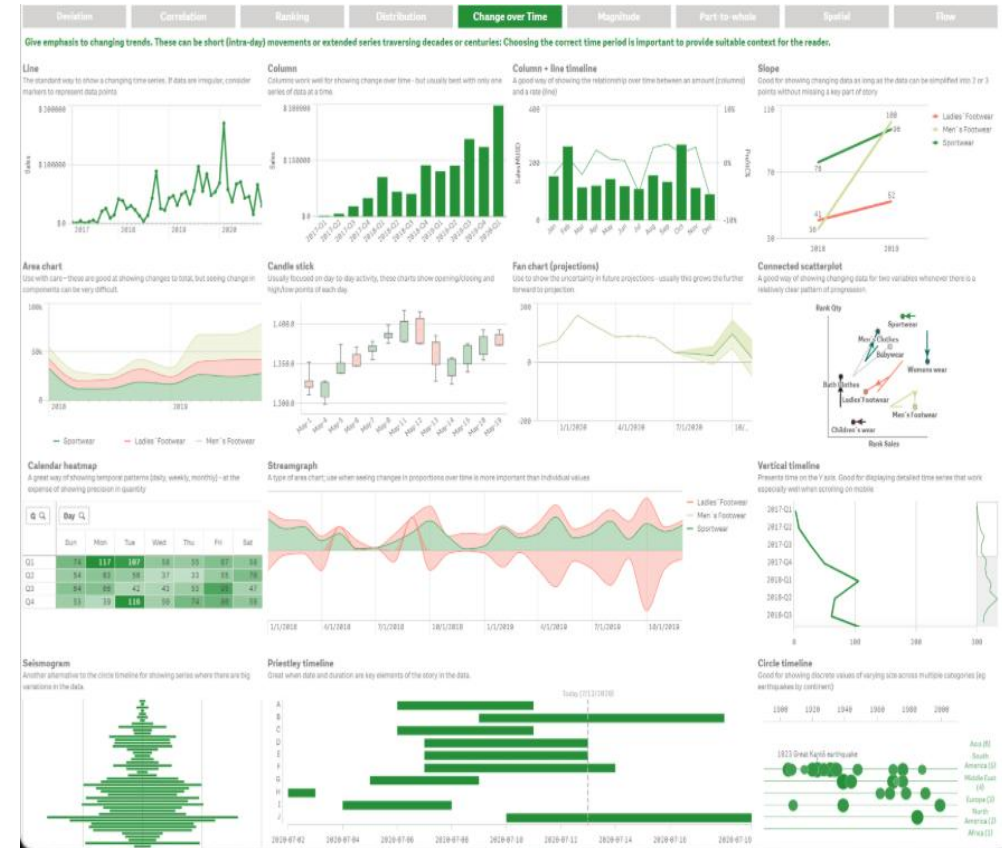


Semantic Analysis

- **Semantic analysis** is the science of extracting meaningful information from speech and textual data.
- Types of semantics analysis:
 - **Natural Language Processing (NLP)**
 - Natural Language Understanding (NLU)
 - Natural Language Generation (NLG)
 - **Text analytics:**
 - It is the process of transforming the unstructured data into meaningful data by applying machine learning, text mining, and NLP techniques
 - It involves:
 - Parsing
 - Searching and retrieval
 - Text mining
 - **Sentiment analysis:**
 - It is analyzing a piece of writing and determining whether it is positive, negative, or neutral.
 - Sentiment analysis is also known as opinion mining

Visual analysis

- **Visual analysis** is the process of analyzing the results of data analysis integrated with data visualization techniques to understand the complex system in a better way
- Various data visualization techniques are used in visual analytics



Data Collection

Data Analysis

Knowledge
extraction

Visualization

**Visual
Analysis**

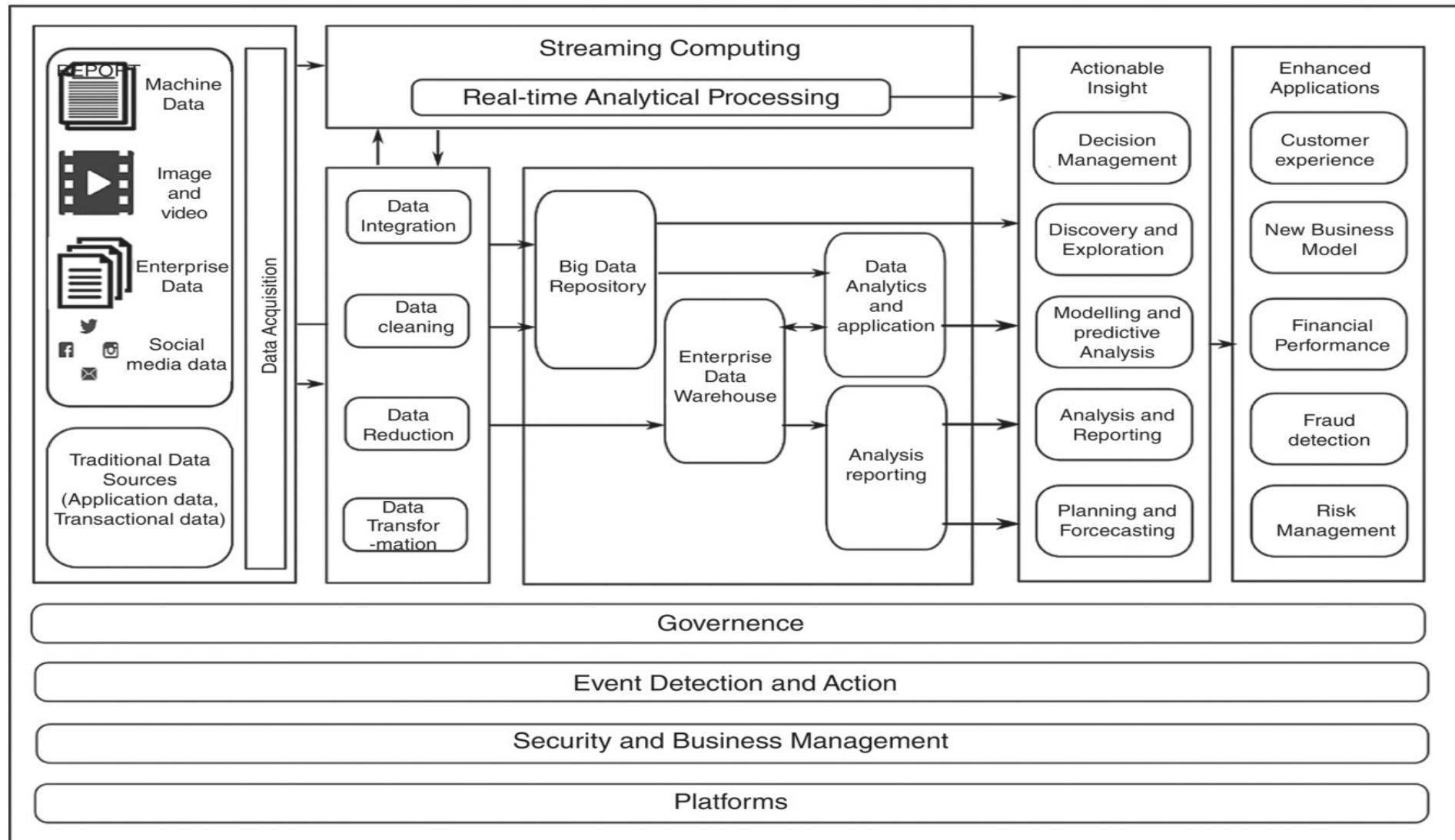
Decision
Making



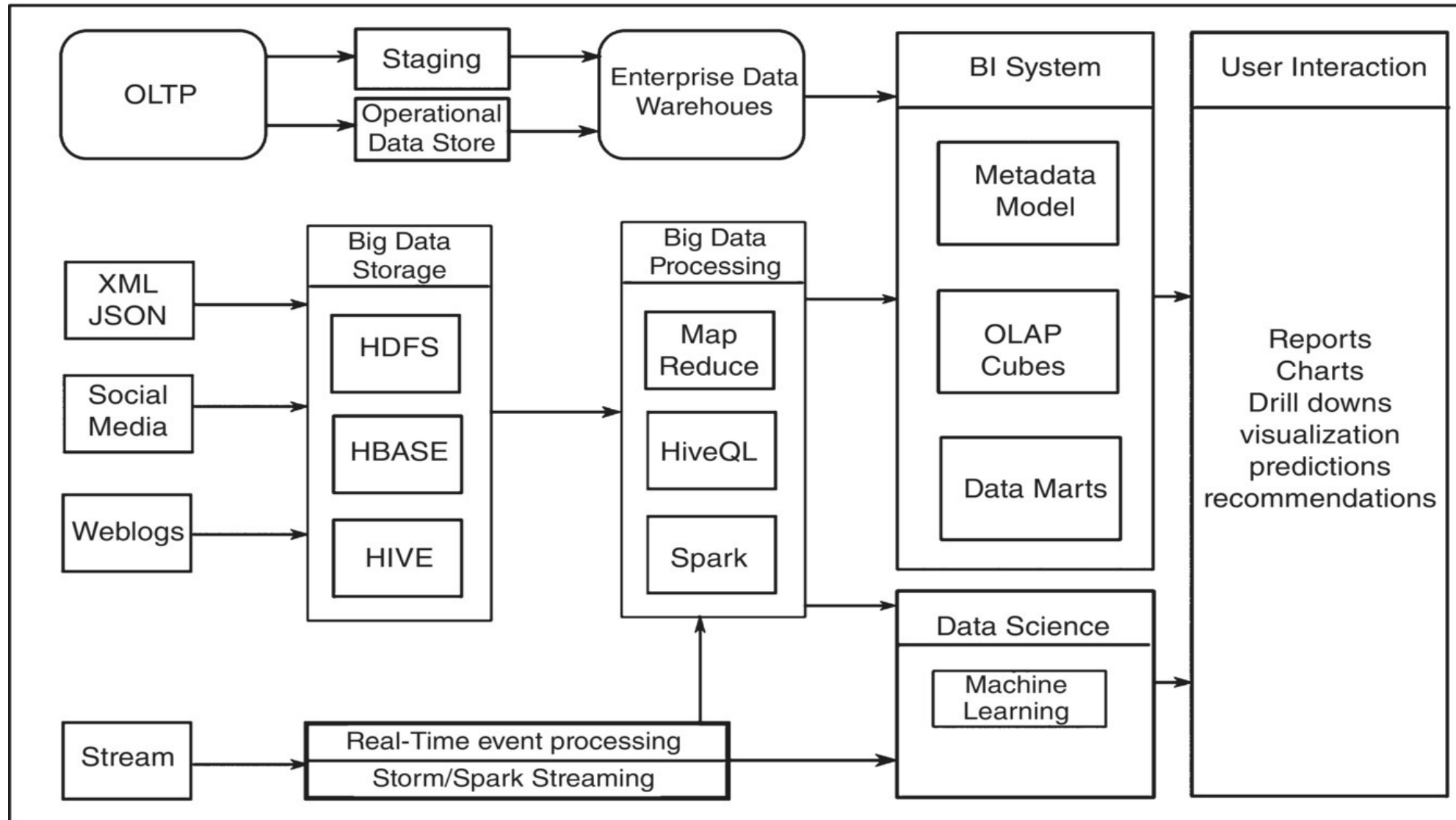
Big Data Business Intelligence

- **Business intelligence (BI)** is the process of analyzing data and producing a desirable output for organizations and end users to assist them in decision-making
 - BI data comprises both data from the storage (previously captured and stored data) and data that is streaming, supporting the organizations to make strategic decisions
- **Online Transaction Processing (OLTP):**
 - Process and manage transaction-oriented applications
 - Examples: airlines, banking, and supermarkets
- **Online Analytical Processing (OLAP):**
 - Process data analysis queries and perform effective analysis on massive amounts of data
 - OLAP is used in forecasting and data mining
 - Examples: prediction of current trends in sales and predicting future prices of commodities
- **Real-Time Analytics Platform (RTAP):**
 - Real-time, in-motion analytics on data from multiple sources at unprecedented speeds and volumes
 - An example patient-monitoring system would alert a doctor or nurse to take a specific action for a situation

Big Data Real-Time Analytics Processing



Integrating Enterprise Data Warehouse with Big Data Technologies



Next Session: Big Data Machine Learning

Q/A

Thank you!